

RTL TO GDS-II WORKSHOP

Project Report on
“RTL to GDS-II implementation of 2^x Logic Generator”



Submitted by:

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Problem Statement:

1. Implement the equation $Y = 2^X$ where X is a 3-bit unsigned number using Verilog.
2. Conditions: -
 - a) Do not use the NOR cell.
 - b) Use the C shape scenario (No. 8) for floor planning.
3. Generate GDS-II layout of design using Synopsys-flowscripts and report the following:
 - a) Design area
 - b) Clock frequency (for a positive setup slack value but less than 1)
 - c) Power consumption

Github Repository Link: https://github.com/Hitanshu22/pow2_3bit

RTL Development

1. RTL Source Code

Filename: pow2_3bit.v

```
`timescale 1ns / 1ps
```

```
module pow2_3bit(
    input clk,
    input rst,
    input [2:0] x,
    output reg [7:0] y
);

    reg [2:0] x_reg;

    always @(posedge clk) begin
        if (rst) begin
            x_reg <= 3'b000;
            y <= 8'b00000000;
        end
        else begin
            x_reg <= x;

            case (x_reg)
                3'd0: y <= 8'b00000001;
                3'd1: y <= 8'b00000010;
                3'd2: y <= 8'b00000100;
                3'd3: y <= 8'b00001000;
                3'd4: y <= 8'b00010000;
```

```
        3'd5: y <= 8'b00100000;
        3'd6: y <= 8'b01000000;
        3'd7: y <= 8'b10000000;
        default: y <= 8'b00000000;
    endcase
end
end

endmodule
```

2. Testbench Code

Filename: pow2_3bit_tb.v

```
`timescale 1ns / 1ps
```

```
module tb_pow2_3bit;

    // Inputs
    reg clk;
    reg rst;
    reg [2:0] x;

    // Output
    wire [7:0] y;

    // Instantiate the DUT (Device Under Test)
    pow2_3bit uut (
        .clk(clk),
        .rst(rst),
```

```
.x(x),  
.y(y)  
);  
  
// Clock generation (10 ns period)  
always #5 clk = ~clk;  
  
// FSDB dump  
initial begin  
    $fsdbDumpfile("pow2_3bit.fsdb");  
    $fsdbDumpvars(0, tb_pow2_3bit);  
end  
  
// Stimulus  
initial begin  
    // Initialize  
    clk = 0;  
    rst = 1;  
    x = 3'b000;  
  
    // Apply reset  
    #10 rst = 0;  
  
    // Apply test inputs  
    #10 x = 3'd0;  
    #10 x = 3'd1;  
    #10 x = 3'd2;  
    #10 x = 3'd3;
```

```
#10 x = 3'd4;
```

```
#10 x = 3'd5;
```

```
#10 x = 3'd6;
```

```
#10 x = 3'd7;
```

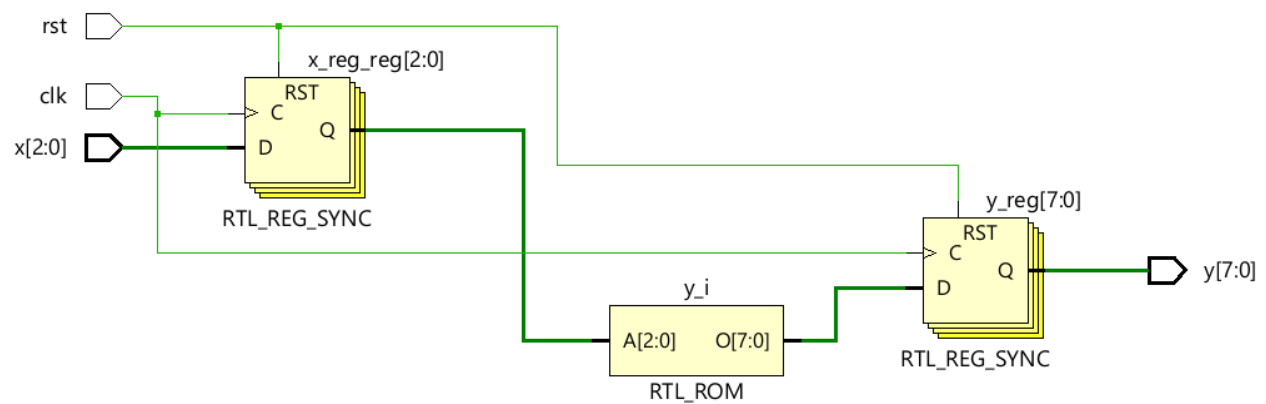
```
// Finish simulation
```

```
#20 $finish;
```

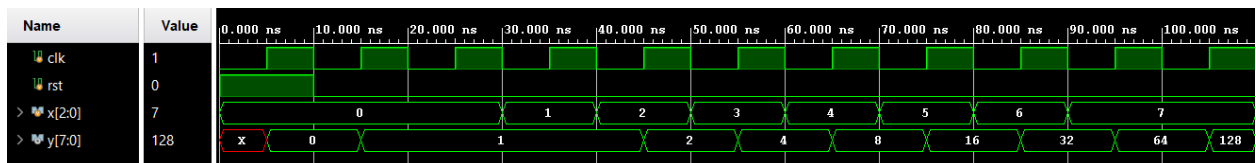
```
end
```

```
endmodule
```

3. RTL Schematic:



4. Timing Diagram:



Logic Synthesis

1. Changes in common_setup.tcl File:

- a) DESIGN_NAME : pow2_3bit

2. Constraint File:

File name: pow2_3bit.sdc

Clock definition

```
create_clock -period 10 -name "clk" [get_ports clk]
```

```
set_clock_uncertainty 0.2 [get_clocks clk]
```

```
set_clock_transition 0.1 [get_clocks clk]
```

Input/Output delays

```
set_input_delay 2.0 -clock clk [all_inputs]
```

```
set_output_delay 2.0 -clock clk [all_outputs]
```

False paths

```
# set_false_path -from [get_ports rst_n]
```

3. Design Compiler TCL commands:

Filename: dc_script.tcl

```
#=====
```

```
# Synopsys Design Compiler
```

```
#=====
```

```
#-----
```

```
# 1. SETUP & ENVIRONMENT
```

```
#-----
```

```
set DESIGN_NAME    "prob1"
set RTL_DIR        "../RTL/"
set CONSTRAINTS_FILE "../constraints/${DESIGN_NAME}.sdc"
set LIBRARY_DIR     "../ref/lib/stdcell_rvt/"
set OUTPUT_DIR      "../outputs"
set REPORTS_DIR     "../reports"
```

Target library (technology .db files)

```
set target_library "$LIBRARY_DIR/saed32rvt_ss0p7vn40c.db"
set link_library   "* $LIBRARY_DIR/saed32rvt_ss0p7vn40c.db
$LIBRARY_DIR/saed32rvt_ff1p16v125c.db"
```

Create output directories

```
file mkdir $OUTPUT_DIR
```

```
file mkdir $REPORTS_DIR
```

#-----

2. READ RTL SOURCE FILES

#-----

Option A: Read Verilog

```
read_verilog [list \
    $RTL_DIR/${DESIGN_NAME}_rtl.v \
]
```

Option B: Read SystemVerilog

```
# read_file -format sverilog [list \
#    $RTL_DIR/top_module.sv \
```


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```
# $RTL_DIR/submodule_a.sv \  
# ]
```

```
# Option C: Read VHDL
```

```
# read_vhdl [list \  
# $RTL_DIR/top_module.vhd \  
# ]
```

```
# Set the top-level design
```

```
current_design $DESIGN_NAME
```

```
#-----
```

```
# 3. LINK THE DESIGN
```

```
#-----
```

```
link
```

```
#-----
```

```
# 4. SET DON'T USE CELLS
```

```
#-----
```

```
#set_dont_use [get_lib_cells */FADD*]
```

```
#set_dont_use [get_lib_cells */HADD*]
```

```
#set_dont_use [get_lib_cells */AO*]
```

```
#set_dont_use [get_lib_cells */OA*]
```

```
#set_dont_use [get_lib_cells */NAND*]
```

```
#set_dont_use [get_lib_cells */XOR*]
```

```
set_dont_use [get_lib_cells */NOR*]
```

```
#set_dont_use [get_lib_cells */XNOR*]
```

```
#set_dont_use [get_lib_cells */MUX*]
```

```
#-----
```

```
# 5. APPLY CONSTRAINTS (SDC)
```

```
#-----
```

```
source $CONSTRAINTS_FILE
```

```
#-----
```

```
# 6. COMPILE (SYNTHESIS)
```

```
#-----
```

```
# Standard compile
```

```
# compile -map_effort medium -area_effort medium
```

```
# Ultra compile
```

```
compile_ultra
```

```
# Incremental compile for further optimization (optional)
```

```
# compile_ultra -incremental
```

```
# Fix hold violations
```

```
#set_fix_hold [all_clocks]
```

```
#compile_ultra -incremental -only_hold_time
```

```
#-----
```

```
# 7. POST-COMPILE OPTIMIZATIONS
```

```
#-----
```

```
# Remove unconnected ports/nets
```

```
remove_unconnected_ports [get_cells -hierarchical *]
```

```
#-----
```

```
# 8. REPORTS
```

```
#-----
```

```
# Timing reports
```

```
report_timing -path full -delay max -max_paths 10 \  
> $REPORTS_DIR/timing_setup.rpt
```

```
report_timing -path full -delay min -max_paths 10 \  
> $REPORTS_DIR/timing_hold.rpt
```

```
# Area report
```

```
report_area -hierarchy \  
> $REPORTS_DIR/area.rpt
```

```
# Power report
```

```
report_power -hierarchy \  
> $REPORTS_DIR/power.rpt
```

```
# Design rule checks
```

```
report_constraint -all_violators \  
> $REPORTS_DIR/constraints.rpt
```

```
report_cell \  
    > $REPORTS_DIR/cells.rpt
```

```
report_net \  
    > $REPORTS_DIR/nets.rpt
```

```
# QoR summary  
report_qor \  
    > $REPORTS_DIR/qor.rpt
```

```
#-----  
# 9. WRITE OUTPUTS  
#-----
```

```
# Remove multi-port nets and constant logic assign statements from netlist  
#set_fix_multiple_port_nets -all -buffer_constants
```

```
# Write gate-level netlist (Verilog)  
write -format verilog -hierarchy \  
    -output $OUTPUT_DIR/${DESIGN_NAME}_netlist.v
```

```
# Write SDC constraints  
write_sdc $OUTPUT_DIR/${DESIGN_NAME}.sdc
```

```
# 10. DONE
```

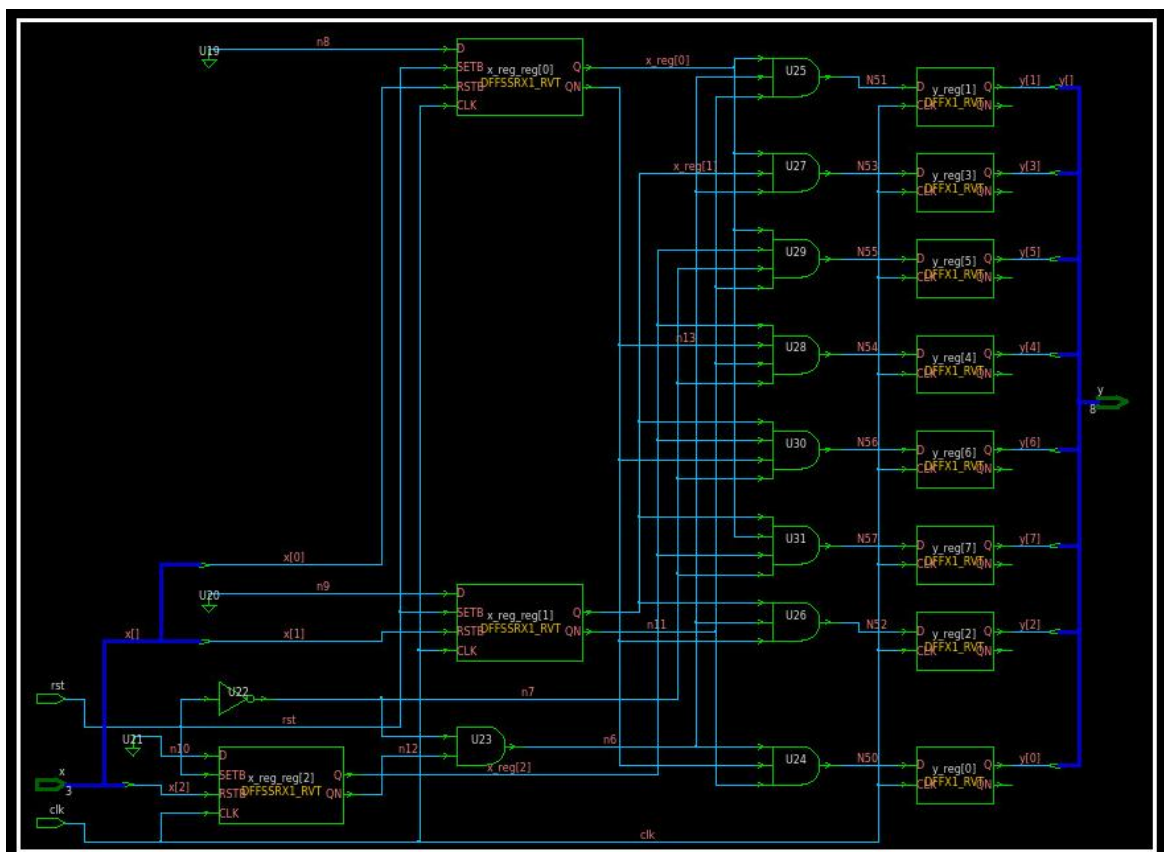
```
echo "=====  
echo " Synthesis COMPLETE: $DESIGN_NAME"
```

Parameters changed: -

- RTL_SOURCE_FILES : “./../rtl_simulation/pow2_3bit.v”
- NOR cells set to don't use [To increase the number of cells in design]

4. Generated outputs:

1. Schematics:



2. Report content:

Category	Parameter	Value
Timing (Hold)	Worst Hold Slack	0.74 ns
Timing (Setup)	Worst Setup Slack	6.55 ns
Area	Total Cells	24
	Combinational Cells	10
	Sequential Cells	11
	Buffers/Inverters	1
	Total Cell Area	96.83 μm^2
	Total Area	103.34 μm^2
Power	Switching Power	0.0645 μW
	Internal Power	3.272 μW
	Leakage Power	0.155 μW
	Total Power	3.492 μW

Physical Design

1. Floor-planning:

#scenario8:

initialize_floorplan -core_utilization 0.6 -coincident_boundary true -core_offset {2 2} -shape U -orientation E

Input pins (clk, rst, x[2:0])

```
set_individual_pin_constraints \  
-ports [get_ports {clk rst x[*]}] \  
-sides {1 2} \  
-pin_spacing_distance 2
```

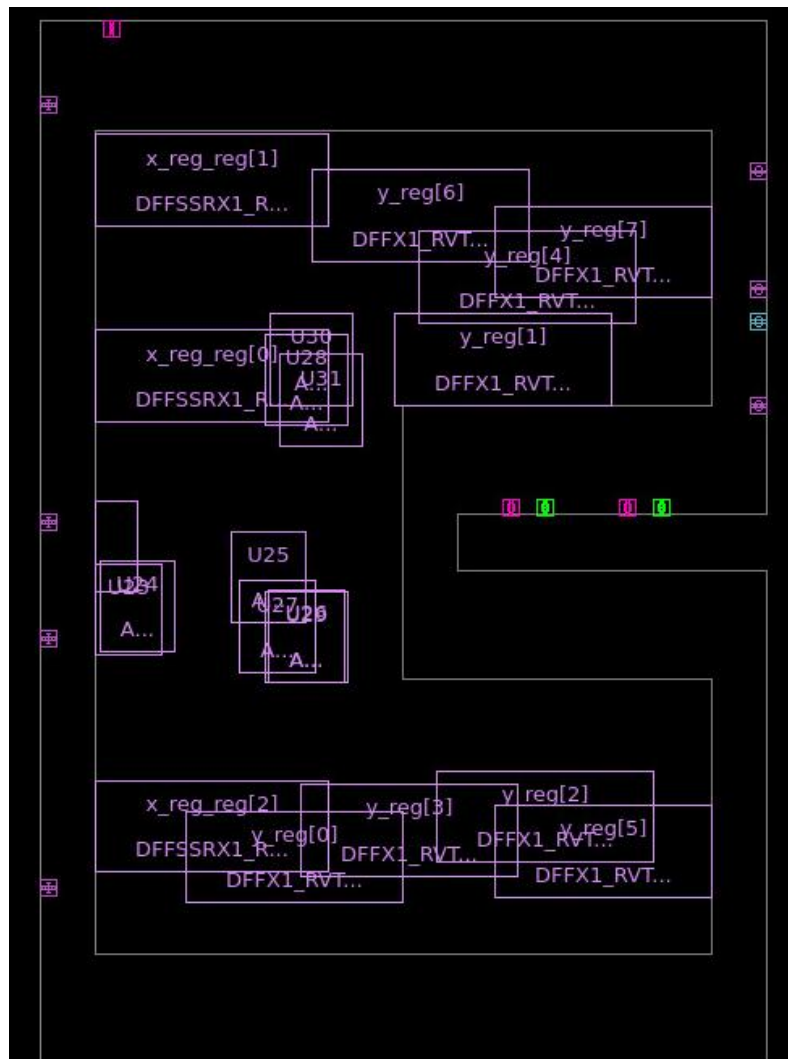
Output pins (y[7:0])

```
set_individual_pin_constraints \  
-ports [get_ports {y[*]}] \  
-sides {3 4} \  
-pin_spacing_distance 2
```

place_pins -self

create_placement -floorplan -effort medium

Design Layout:



2. Power planning:

#-----

4b. Power / ground rings

#-----

```
create_pg_ring_pattern core_ring_pattern -horizontal_layer M7 -horizontal_width .4 -
horizontal_spacing .3 -vertical_layer M8 -vertical_width .4 -vertical_spacing .3
```

```
set_pg_strategy core_power_ring -core -pattern {{name : core_ring_pattern} {nets : {VDD
VSS}}} {offset : {.5 .5}}}
```

```
compile_pg -strategies core_power_ring
```



```
#-----  
# 4c. Power mesh  
#-----  
  
create_pg_mesh_pattern mesh -layers { {{vertical_layer: M6} {width: .34} {spacing:  
interleaving} {pitch: 5} {offset: .5}} {{horizontal_layer: M7} {width: .38} {spacing:  
interleaving} {pitch: 5} {offset: .5}} {{vertical_layer: M8} {width: .38} {spacing:  
interleaving} {pitch: 5} {offset: .5}}} }  
  
set_pg_strategy core_mesh -pattern { {pattern:mesh} {nets: VDD VSS}} -core -extension {stop:  
innermost_ring}  
  
compile_pg -strategies core_mesh  
  
#-----  
# 4d. Standard-cell power rails  
#-----  
  
create_pg_std_cell_conn_pattern std_cell_rail -layers {M1} -rail_width 0.06  
  
set_pg_strategy rail_strat -core -pattern {{name: std_cell_rail} {nets: VDD VSS} }  
  
compile_pg -strategies rail_strat
```

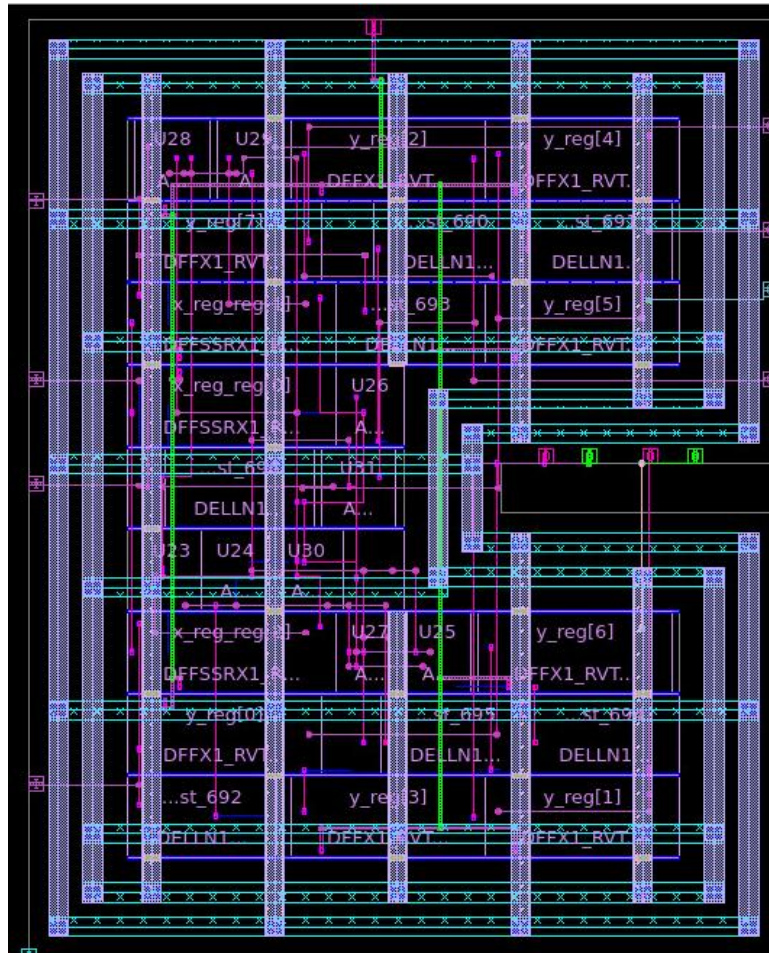
3. Placement:

Set NOR cells to don't use [Similar constraint during Logic Synthesis].

```
#set_dont_use [get_lib_cells */FADD*]  
#set_dont_use [get_lib_cells */HADD*]  
#set_dont_use [get_lib_cells */AO*]  
#set_dont_use [get_lib_cells */OA*]  
#set_dont_use [get_lib_cells */NAND*]  
#set_dont_use [get_lib_cells */XOR*]  
set_dont_use [get_lib_cells */NOR*]  
#set_dont_use [get_lib_cells */XNOR*]  
#set_dont_use [get_lib_cells */MUX*]
```

4. Clock Tree synthesis:

Design layout:



5. Routing:

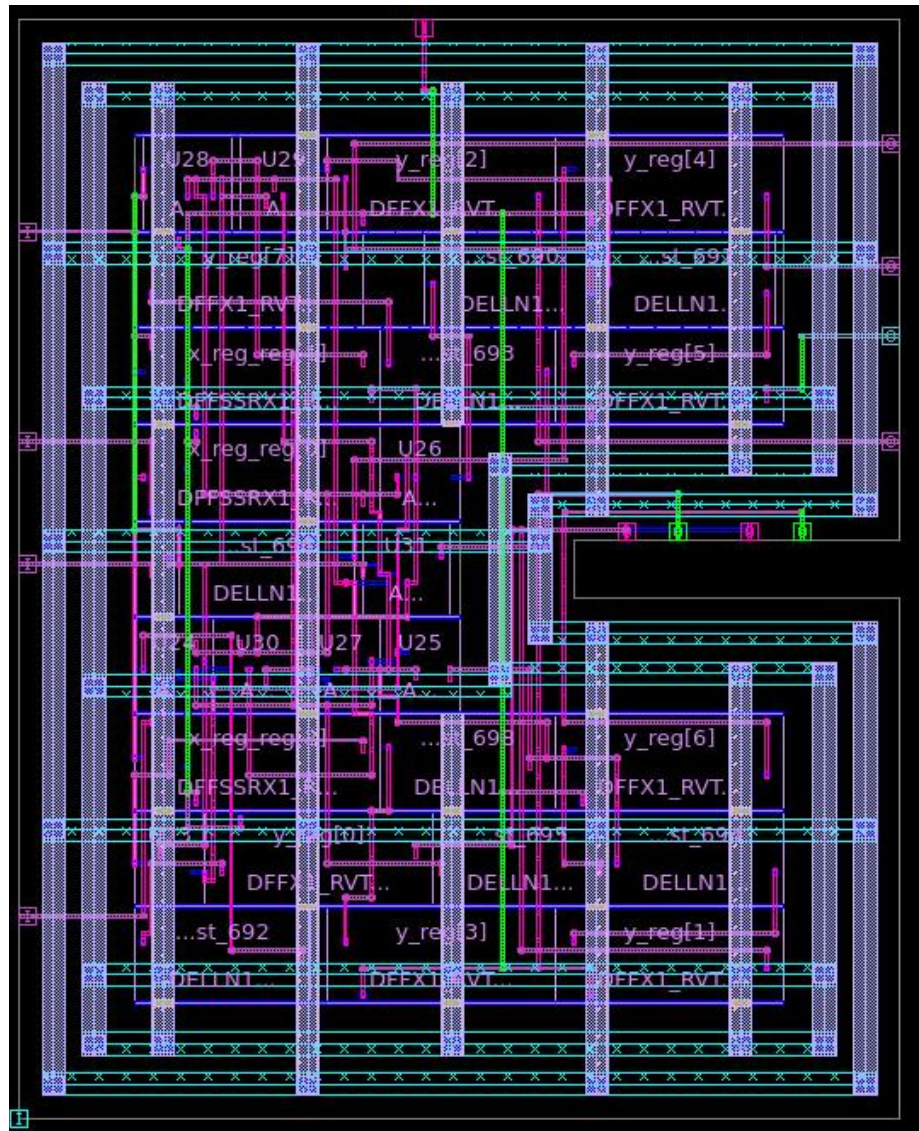
7a. Global route options

```
set_app_options -name route.global.timing_driven -value true
```

```
set_app_options -name route.global.crosstalk_driven -value false
```

```
#-----  
# 7b. Track assignment options  
set_app_options -name route.track.timing_driven -value true  
set_app_options -name route.track.crosstalk_driven -value true  
#-----  
# 7c. Detail route options  
set_app_options -name route.detail.timing_driven -value true  
#set_app_options -name route.detail.save_after_iterations -value false  
set_app_options -name route.detail.force_max_number_iterations -value false  
set_app_options -name route.detail.antenna -value true  
set_app_options -name route.detail.antenna_fixing_preference -value use_diodes  
set_app_options -name route.detail.diode_libcell_names -value */ANTENNA_RVT  
#-----  
# 7d. Atomic routing sequence  
route_global  
route_track  
route_detail  
#-----  
# 7e. Post-route optimisation  
#-----  
  
route_opt
```

Final IC Layout after all the Physical design flow:



Post Physical Design report file analysis**1. Floorplanning:**

Slack	0.74 ns
Power consumption	3.492 μ W
Area	103.34 μ m ²

2. Powerplanning and Placement:

Slack	0.74 ns
Power consumption	3.492 μ W
Area	103.34 μ m ²
Connectivity	Fully connected

3. Clock Tree Synthesis:

Slack	0.74 ns
Power consumption	3.492 μ W
Area	103.34 μ m ²
Clock latency	~0.07 ns (approx.)

4. Routing:

Slack	0.74 ns
Power consumption	3.492 μ W
Area	103.34 μ m ²
PG Connectivity	Fully connected
Routing Verification	No open nets

Static Timing Analysis

1. Late corner:

```
set report_default_significant_digits 6
set link_path "../ref/lib/stdcell_rvt/saed32rvt_ff1p16v125c.db"

set DESIGN_NAME "pow2_3bit"
set REPORTS_DIR "./reports"
file mkdir $REPORTS_DIR

read_verilog "../ICCII/outputs/${DESIGN_NAME}.routed.v"
link_design
current_design $DESIGN_NAME

read_sdc "../ICCII/outputs/${DESIGN_NAME}_final.sdc"

# Late corner SPEF (slow)
read_parasitics "../ICCII/outputs/${DESIGN_NAME}_func::nom.spef.p1_125.spef"

update_timing -full

report_timing -path full -delay max -max_paths 10 \
    > $REPORTS_DIR/timing_setup.rpt

report_design > $REPORTS_DIR/design_setup.rpt
```

```
*****
Report : timing
        -path_type full
        -delay_type min
        -slack_lesser_than 0.000000
        -max_paths 10
        -sort_by slack
Design : pow2_3bit
Version: W-2024.09
Date   : Sun Apr  5 15:38:53 2026
*****

No paths with slack less than 0.000000.
```

2. Early corner:

```
set report_default_significant_digits 6
set link_path "../ref/lib/stdcell_rvt/saed32rvt_fflp16v125c.db"

set DESIGN_NAME "pow2_3bit"
set REPORTS_DIR "./reports"
file mkdir $REPORTS_DIR

read_verilog "../ICCII/outputs/${DESIGN_NAME}.routed.v"
link_design
current_design $DESIGN_NAME

read_sdc "../ICCII/outputs/${DESIGN_NAME}_final.sdc"

# Early corner SPEF (fast)
read_parasitics "../ICCII/outputs/${DESIGN_NAME}_func::nom.spef.p2_125.spef"
```



```
update_timing -full
```

```
report_timing -path full -delay min -max_paths 10 \  
> $REPORTS_DIR/timing_hold.rpt
```

```
report_design > $REPORTS_DIR/design_hold.rpt
```

```
*****  
Report : timing  
        -path_type full  
        -delay_type max  
        -slack_lesser_than 0.000000  
        -max_paths 10  
        -sort_by slack  
Design : pow2_3bit  
Version: W-2024.09  
Date   : Sun Apr  5 15:38:53 2026  
*****  
  
No paths with slack less than 0.000000.
```

3. Report Content:

Slack	6.55 ns (setup), 0.74 ns (hold)
Violations	None
Timing Status	MET

Final Results

Cell Area	96.83 μm^2
Total Area	103.34 μm^2
Power Consumption	3.492 μW
Clock Frequency	~100 MHz (10 ns clock)

Learnings

I. RTL Development:

- To implement a circuit using Behavioural modelling in Verilog HDL
- To simulate a Verilog testbench using VCS and Verdi tools

II. Logic Synthesis

- Using the Design Compiler tool to build the gate-level logic design of the provided Verilog module • Setting up the constraints and updating constraints to adjust the outputs
- Understood the concepts like worst setup, slack, and process corners, which create an impact on our design

III. Physical Design.

- Learned to create a floorplan and power plan for the physical design using the IC Compiler tool, and to place cells based on the power plan
- Learned to check whether the design is connected or not to verify the above steps
- Learned to synthesize a clock tree and check its latency
- Learned the routing process to connect each path physically in the design

IV. Static Timing Analysis

- Using the Primetime tool to perform the Static Timing Analysis of the design
- Learned to interpret the reports generated by tools for the analysis of design
- Learned to implement the VLSI Design Flow from RTL Development to STA with the help of Synopsys Tools